KASCO

Kalimantan SAF Supply Chain Optimiser

AI-Driven Optimisation for a More Sustainable Aviation Future

Technical report, Airbus Challenge 3, Autonomous SG 2026

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Live application: weewzr.github.io/airbustrack3ray. Source: github.com/weewzr/airbustrack3ray.

1. Executive summary

A SAF network planner must place about USD 4.7B of plant capital across Kalimantan before engineering starts. This project gives that planner a tool that searches refinery and depot locations and sizes, routes biomass and SAF exactly, and shows the checks behind each number.

On the official inputs of the 31 October update, the best network found costs **USD 890.09M per year**, or **USD 742 per tonne SAF** before feedstock. It uses 6 large refineries, 23 depot modules (21 large, 2 small), 6,000 kt of biomass from 71 sites, and meets all 1,200 kt of demand. On candidate sites the cost is proven within 0.75% of the optimum. The final coordinates are a local refinement with no global proof.

Refinery cost dominates the decision, since transport is only 7.5% of the total. Under the official inputs depots belong at the refinery, and a HiGHS run with depots free to go anywhere confirmed this on the base fleet. Field depots cost USD 22.6M more per year. They move 18.8% more tonne-km but need an estimated 25% fewer truck-km, a trade-off left for the planner to weigh.

The network costs USD 61.3M to USD 76.7M per year less than two simple heuristic layouts. A strong k-means benchmark, handed the same six-plant fleet and our refinement, reaches our network at best and sits USD 1.1M above it at median. That supports the result and also shows the objective is flat near the optimum. KASCO's value is the proven bound, the fleet and size choice in each scenario, and limits such as the cost step at 1,304.4 kt of SAF.

Reference Frame

Variables and parameters

Symbol	Definition	Units	Value or range
i, j, k, a, s	Biomass site, depot site, refinery site, airport, size (small, large)		80 sites, 4 airports
x_{ij}	Biomass from site i to depot at j	kt/yr	Decision, continuous
u_{jk}	Pre-processed biomass, depot j to refinery k	kt/yr	Decision, continuous
w_{ka}	SAF from refinery k to airport a	kt/yr	Decision, continuous
y_{js}	Depot modules of size s at j	count	Decision, integer
z_{ks}	Refinery of size s open at k	binary	Decision
t_{js}, r_{ks}	Throughput by size at depot and refinery	kt/yr	Decision, continuous
B_i	Biomass available at site i	kt/yr	22.84 to 99.90, sum 6,800
D_a	SAF demand at airport a	kt/yr	300 each
η, ρ	Refinery yield, depot mass retention	t/t	0.20, 1.00
C^D_s, C^R_s	Depot and refinery biomass intake capacity	kt/yr	150, 300; 500, 1,087
F^D_s, F^R_s	Depot and refinery CapEx	USD M	2.25, 4.5; 480, 765
v^D_s, v^R_s	Variable OpEx on biomass in	USD/t	5, 4; 60, 55
τ_1, τ_2, τ_3	Tariff by leg	USD/t/km	0.10, 0.12, 0.08
n, f	CapEx life, fixed OpEx share	yr, 1/yr	20, 0.05
d	Distance between two points	km	Great-circle x 1.3

Equations

The objective is equation (1). Equations (2) to (8) are its constraints. Equation (9) defines distance. Equations (10) to (12) are the solution method. All appear in full in section 6.

Assumptions needed to read them

One-year cost basis with straight-line CapEx and no interest. Biomass mass is taken as stated, with moisture ignored by instruction. All values are deterministic. Masses are annual. Refinery capacity is biomass intake.

2. Problem and industrial context

The named user is a SAF network planner at a project developer or fuel supplier, recommending sites before front-end engineering. Airbus's Singapore office was reported in August 2025 to be working with IPB University on Indonesian biomass availability, logistics, supply chain mapping and factory siting to cut transport cost, with empty fruit bunches in Sumatra and Kalimantan named as feedstock [19]. The same reports list weak infrastructure, an unclear legal framework and limited large-scale processing as obstacles. Those are secondary reports, and their details differ.

We found no published description of how planners work today. Comparing a few layouts in GIS and spreadsheets is our reasoned assumption, and it should be read as such. Mistakes are persistent, since a large refinery fixes USD 765M of capital and its haulage for 20 years. Project risk is real: a gasification Fischer-Tropsch plant in Nevada entered bankruptcy in 2024 after equipment and liquidity problems [20].

SAF supply is scarce, at about 1.9 Mt in 2025 or 0.6% of jet fuel [21]. Indonesia's first requirement, 1% from 2027, covers international flights from its two busiest airports [16], and Pertamina's first SAF was used cooking oil co-processed at Cilacap [22]. Regional studies support residues as a feedstock: about 45.7 Mt of SAF potential across Southeast Asia by 2050 [23], and Fischer-Tropsch among the most viable routes [24]. A preliminary national estimate puts palm residues near 77% of Indonesian waste-based SAF potential [25]. We found no Kalimantan-wide residue tonnage, so the 6,800 kt remains a challenge input.

The physical constraints are real. Biomass clusters in West, Central and East Kalimantan, the airports sit far apart, and Tarakan is an island. A sample of 854 site pairs on OpenStreetMap gave a median road distance of 1.9 times great-circle, above the official 1.3. Older work on Central Kalimantan palm freight found road and river close to evenly used, with intermodal routes the cheapest generalised option [26].

The platform helps that planner decide where to open plants, what size to build, which biomass sites to contract, how to route fuel, and when demand growth forces new capacity. Likely users are SAF producers, fuel distributors, airlines, airport operators, engineering consultancies and government planners. These are possible users, not existing customers.

3. Official Airbus challenge requirements

The brief asks for a SAF supply chain across Kalimantan that minimises OpEx, CapEx and logistics while meeting demand at four airports: Supadio (PNK), Iskandar at Pangkalan Bun (PKN), Sepinggan (BPN) and Juwata (TRK). The chain runs biomass site, pre-processing depot, refinery, airport. The answer must give coordinates, facility sizes and routes. The update of 31 October at 12:45 fixed every open parameter and says it overrides the original where they differ.

4. Data and assumptions

Validation of the official data

Check	Result
Biomass rows and total	80 rows, 6,800.00 kt, matching the brief
Range per site	22.84 to 99.90 kt, inside the stated 20 to 100 kt
Null cells, duplicate ids	0, 0
kt and tonnes columns agree	Yes, on every row
Airports	4 rows, 300 kt each, 1,200 kt total
Refinery cost scaling	765 / 480 equals $(1{,}087 / 500)^{0.60}$, the six-tenths rule
Depot unit CapEx	USD 15k per kt for both sizes, so large wins only on variable cost
Coastline check	13 of 80 points, 1,153 kt, fall off the Natural Earth 10 m coast

The 80 points should be read as modelling inputs, not surveyed production sites. 12 of the 13 off-coast points are labelled Ketapang, and the farthest lies 37.5 km from the coastline, too far to be a mapping error. We keep them as biomass sources because the coordinates are official, but they cannot host a refinery since the brief requires land. The first solver version missed this, and the independent audit caught a plant placed on one of those points.

The script `check_official.py` compares 30 engine inputs with the official workbook and airport file. All match. Real-uplift figures come from `fetch_context.py`, which fetches each source page and records whether the quoted figure was found. They are used only in the labelled reality check.

Assumption register

Input	Value	Status	Source
Capacity basis	Biomass intake	Official	Update, 12:45
Yield	0.20 t SAF per t biomass	Official	Update, 12:45
Depot mass loss	0%	Official	Update, 12:45
Demand	300 kt per airport	Official	challenge3_airports_4.csv
Costs	CapEx / 20, fixed OpEx 5%, variable per tonne	Official	challenge3_parameters.xlsx
Distance	Great-circle x 1.3	Official	Update, 12:45
Tarakan leg	Barge at 0.08 USD/t/km	Official	Update, 12:45
Refinery candidates	On-land sites plus PNK, PKN, BPN	Ours	Tarakan excluded, as biomass cannot reach the island
One refinery per site	At most one	Ours	Depot modules may share a site
Field depot radius	60 km	Ours	Used only in the field design
Real uplift	About 100 kt/yr	Public	Reality check only, see section 13

5. Literature review

Six papers shaped the formulation. Ng and Maravelias [3] build a multi-period MILP with variable depot and biorefinery locations, and report that sites near a biorefinery ship directly while distant sites justify a depot. That pattern depends on densification lowering later transport cost, which the brief's tariffs do not offer. Their example took 3,405 s to reach a 0.43% gap with 431 binaries, a warning about scale. Ng et al. [2] stress accurate distances and show that pruning reduces 557 candidate depots and over three million arcs to 378 depots and 80,000 arcs, which supports candidate reduction.

Huang et al. [5] compare ATJ, FT and HTL on corn stover. They report that large centralised plants win through economies of scale, which matches our result. Li et al. [4] show that assuming fixed yields overstates maximum SAF supply by 14.15% and net revenue by 18.27% when ash varies, so yield deserves scrutiny. Hernandez-Araujo et al. [6] couple supply chain and process models for bagasse-based ATJ, and Castillo-Villar et al. [7] use machine learning to pick facility configurations for a county, depot, biorefinery, airport network, which is the same four-tier shape as this problem.

We take the four-tier mass balance from [7], the strong linking constraints from the facility location literature, and the candidate-reduction idea from [2]. We did not combine their cost data, because the update fixes all costs. Course material from CN4252 [8, 9] informed the annualisation logic and the assumption register.

Three papers shaped what we chose not to build. Ng et al. [2] support real road distances, which we test as a sensitivity because the official distance is fixed. Li et al. [4] warn against fixed yields. Their figures come from US logging residues and do not transfer, but as an illustration only, a 14% shortfall on 0.20 would give 0.172, below our feasibility threshold of 0.176. Keyvandarian et al. [27] apply adaptive robust optimisation to SAF allocation under production uncertainty. We lack data to calibrate an uncertainty set, so that stays future work, and the official 0.20 stays unchanged.

6. Mathematical formulation

Minimise annual cost:

$$\min\ (\tfrac{1}{n}+f)\Big[\sum F_s^D y_{js}+\sum F_s^R z_{ks}\Big]+\sum v_s^D t_{js}+\sum v_s^R r_{ks}+\sum \tau_1 d_{ij}x_{ij}+\sum \tau_2 d_{jk}u_{jk}+\sum \tau_3 d_{ka}w_{ka} \quad (1)$$

Subject to biomass availability, depot balance and capacity, depot mass retention, refinery balance and capacity, conversion, demand, and domains:

$$\sum_j x_{ij} \le B_i \quad \forall i \quad (2)$$

$$\sum_i x_{ij} = \sum_s t_{js}, \qquad t_{js} \le C_s^D\, y_{js} \quad \forall j,s \quad (3)$$

$$\sum_k u_{jk} = \rho \sum_i x_{ij} \quad \forall j \quad (4)$$

$$\sum_j u_{jk} = \sum_s r_{ks}, \qquad r_{ks} \le C_s^R\, z_{ks} \quad \forall k,s \quad (5)$$

$$\sum_a w_{ka} = \eta \sum_s r_{ks} \quad \forall k \quad (6)$$

$$\sum_k w_{ka} \ge D_a \quad \forall a \quad (7)$$

$$\sum_s z_{ks} \le 1, \quad z_{ks} \in \{0,1\}, \quad y_{js} \in \mathbb{Z}_{\ge 0}, \quad x,u,w,t,r \ge 0 \quad (8)$$

$$d_{pq} = 1.3 \times 2R \arcsin\sqrt{\sin^2\tfrac{\Delta\phi}{2} + \cos\phi_p \cos\phi_q \sin^2\tfrac{\Delta\lambda}{2}} \quad (9)$$

Truck size

The tariff is per tonne-km and the same for both truck sizes, so truck size cannot enter the objective. We report trip counts only, and we do not model trucks as a decision.

Why depots sit at refineries

Take biomass moving from site i through a depot at j to a refinery at k. Its transport cost per tonne is $\tau_1 d_{ij} + \rho\tau_2 d_{jk}$. If $\rho\tau_2$ is at least τ_1, this is at least $\tau_1(d_{ij} + d_{jk})$, which by the triangle inequality is at least $\tau_1 d_{ik}$. For a depot serving one refinery, moving it to that refinery therefore never raises transport cost and keeps the same modules.

A depot feeding two refineries needs more care, since splitting it could add a module. The argument alone therefore leaves a margin, bounded by the difference between actual depot cost and its linear floor: USD 1.03M per year on the base fleet.

We closed that margin computationally. With the six base-case refineries fixed, HiGHS solved the full three-tier model with depots at any site, shared between refineries, in integer modules. It proved USD 890.3215M (0.0000% gap), the same as the co-located cost, with 0 shared and 0 remote depots.

The result needs metric distances, $\rho\tau_2$ at or above τ_1, and this fleet. It reverses below a retention of 0.833, which is why the instruction to ignore moisture loss matters. A real depot that dries wet biomass could earn its place in the field, so this is a result about these inputs, not a general rule.

7. Solver architecture

Method	Assessment	Decision
General MILP solver (HiGHS)	Reliable with a proven bound, but C++ and heavy to run live in a browser	Kept as the independent check
Purpose-built Lagrangian branch-and-bound in Rust	Exploits problem structure, compiles to WebAssembly, gives a true bound	Chosen engine
MINLP with free coordinates	Non-convex, no practical global proof at this size	Replaced by a two-step method
Clustering plus MILP	Fast but depends on arbitrary cluster counts	Used only as a starting design
Pure metaheuristic	No bound, so no honest statement of quality	Rejected

Measurements drove this choice. In a timed run of a general MILP on an earlier version of this problem, 99.1% of runtime sat inside the solver, so a language change alone could not help. The gain came from using the structure. Relaxing supply and demand with multipliers u and λ splits the model into one subproblem per site:

$$L(u,\lambda) = \sum_a \lambda_a E_a - \sum_i u_i B_i + \min_z \sum_{k,s} z_{ks}\,(G_s + V_{ks}(u,\lambda)), \qquad \sum_{k,s} C_s^R z_{ks} \geq \sum_a E_a \tag{10}$$

Here E_a is the biomass equivalent of demand, G_s the annual fixed cost of a plant and V_{ks} the best value of site k, found by pairing the cheapest inputs with the dearest outputs. Any multipliers give a valid lower bound, and subgradient steps tighten it. Each design found is priced exactly by min-cost flow.

The relaxed problem may pick one site in both sizes. This keeps the bound valid, because it only removes a constraint, and it occurred in 0.6% of relaxed solutions on the base case. Reported designs are always one size per site. Branching adds little on this instance: biomass sites are near-substitutes, so closing one barely moves the bound, and the reported gap is close to the root relaxation. A best-first variant tested outside the repository did not improve it.

Sites may be anywhere on land, so a final step alternates a location move and a re-allocation [10], where each plant moves by Weiszfeld iteration [11]:

$$\mathbf{p}^{(t+1)} = \left(\sum_m \frac{\omega_m \mathbf{p}_m}{\|\mathbf{p}^{(t)} - \mathbf{p}_m\|}\right)\left(\sum_m \frac{\omega_m}{\|\mathbf{p}^{(t)} - \mathbf{p}_m\|}\right)^{-1}, \qquad \omega_m = \tau \times \mathrm{flow}_m \tag{11}$$

A land mask rejects offshore moves. This step converges to a local optimum and carries no global proof.

System architecture

The engine is a Rust library covering data, instance building, min-cost flow, search, refinement, audit, baselines, diagnostics and decision support, with 38 automated tests. A command-line tool runs batches and

exports the model. A WebAssembly build of the same library runs inside the browser in a Web Worker, so the deployed site on GitHub Pages needs no server, and the solver needs no account or API key. The page uses Leaflet for the map and Chart.js for charts. Python appears only in supporting scripts for data fetching, the HiGHS check and these documents.

8. Optimisation results

Measure	Value
Total annual cost	USD 890.09M
Cost per tonne SAF	USD 742
Candidate-site cost	USD 890.32M
Proven lower bound, candidate sites	USD 883.60M
Proven gap	0.75%
Search effort	3,975 nodes, 298,958 bound iterations, 60 s
Biomass used	6,000 kt from 71 sites
SAF delivered	1,200 kt, 100% of demand
HiGHS cost, same model	USD 890.32M (Time limit reached)
HiGHS lower bound	USD 866.74M
HiGHS gap and time limit	2.65%, 480 s

9. Geographic network results

Refinery	Size	Latitude	Longitude	Biomass kt	SAF kt	Used	Nearest site
R1	large	-2.0113	116.0324	807	161	74.2 %	24 km from KAL_SUP_24 (Paser, East Kalimantan)
R2	large	-1.7167	110.0022	1,087	217	100.0 %	1 km from KAL_SUP_41 (Ketapang, West Kalimantan)
R3	large	-0.3788	116.9294	986	197	90.7 %	4 km from KAL_SUP_44 (Kutai Kartanegara, East Kalimantan)
R4	large	0.2508	110.6944	1,087	217	100.0 %	3 km from KAL_SUP_46 (Sanggau, West Kalimantan)
R5	large	-2.5351	111.6833	1,087	217	100.0 %	2 km from KAL_SUP_65 (Kotawaringin Barat, Central Kalimantan)
R6	large	-2.4215	112.9479	947	189	87.1 %	1 km from KAL_SUP_71 (Kotawaringin Timur, Central Kalimantan)

Depot modules at each refinery: R1: 3 large; R2: 4 large; R3: 1 small, 3 large; R4: 4 large; R5: 4 large; R6: 1 small, 3 large.

Continuous refinement saved only USD 0.23M per year. Biomass sites sit in tight clusters, so the best candidate site is already close to the geometric median. As a stress test we added 510 land grid points at 0.2 degree spacing to the 70 candidates. After refinement that run reached USD 890.09M, which is strong evidence, though not proof, that the candidate restriction hides little.

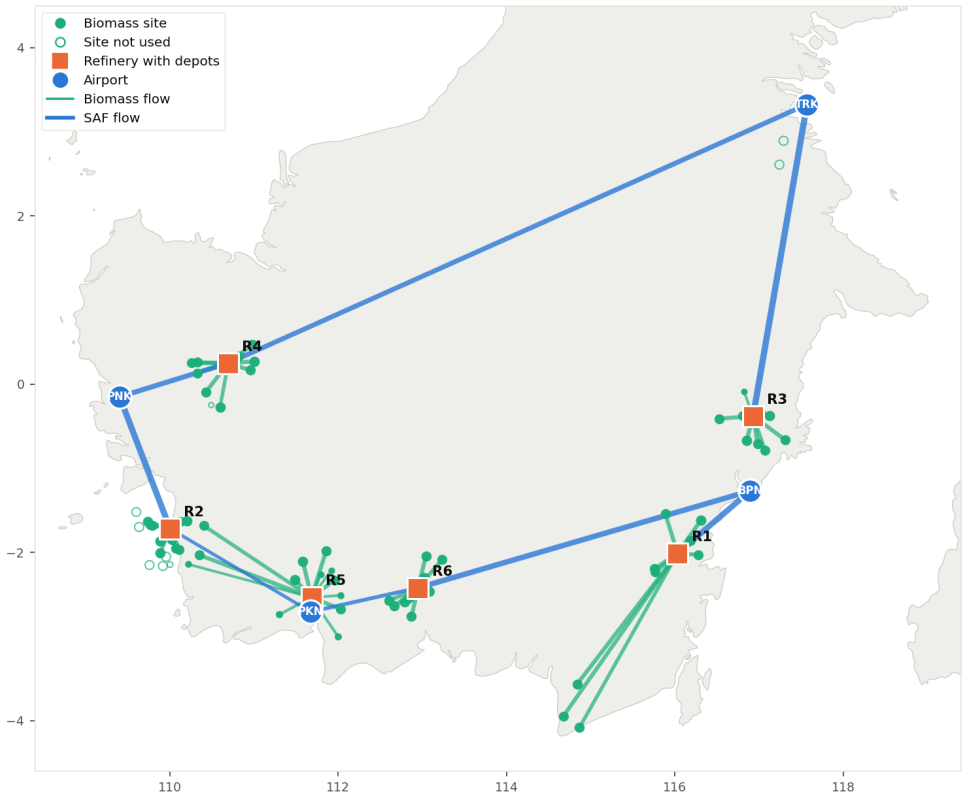


Figure 1. Optimised network. Refineries sit inside the biomass clusters, with depots at each refinery. Lines are schematic, and the last leg to Tarakan is by barge.

Preliminary site screening

Preliminary. OpenStreetMap coverage is uneven, so a missing feature proves nothing. No site is shown to be buildable. Land tenure, peat and forest status are unchecked. The table adds no constraint to the optimiser, and data come from OSRM and OpenStreetMap within 25 km.

Plant	To road, m	Named rivers	Ports, piers	Industrial	Protected nearby
R1	614	Sungai Apar Besar, Sungai Kendilo, Sungai Kerang, Sungai Kesunge	0	2	Cagar Alam Teluk Adang
R2	537	Sungai Ketapang, Sungai Pawan	6	1	None mapped
R3	7	Sungai Jembayan, Sungai Mahakam, Sungai Tenggarong	56	5	None mapped
R4	148	Sungai Engkatat, Sungai Engkode, Sungai Kantu, Sungai Kapuas	7	6	None mapped
R5	1,428	Sungai Arut, Sungai Kumai, Sungai Lamandau	28	5	unnamed
R6	837	Sungai Baamang, Sungai Cempaga, Sungai Mentawa, Sungai Mentaya	17	6	None mapped

Transport and material flows

From	To	SAF kt	Distance km	USD M/yr	Mode
R1	BPN	161.3	164	2.12	truck
R2	PNK	185.5	242	3.60	truck
R2	PKN	31.9	280	0.72	truck
R3	TRK	197.1	544	8.57	truck then barge
R4	PNK	114.5	195	1.79	truck
R4	TRK	102.9	1,088	8.95	truck then barge
R5	PKN	217.4	25	0.43	truck
R6	PKN	50.7	189	0.76	truck
R6	BPN	138.7	594	6.59	truck

Sites unused: KAL_SUP_03, KAL_SUP_18, KAL_SUP_22, KAL_SUP_31, KAL_SUP_40, KAL_SUP_50, KAL_SUP_69, KAL_SUP_70, KAL_SUP_80. These account for the 800 kt not needed. Mean biomass haul is 56 km and mean SAF haul 349 km. Biomass is five times heavier per tonne of SAF, so plants stay close to it and SAF travels further. Delivery to Tarakan is the longest and the sole barge leg.

Leg	Tonnes	Small truck	Trips	Large truck	Trips
Farm to depot	6,000,000	3 t	2,000,001	9 t	666,667
Depot to refinery	6,000,000	18 t	333,334	30 t	200,000
Refinery to airport	1,200,000	16 t	75,000	32 t	37,500

10. Cost and facility analysis

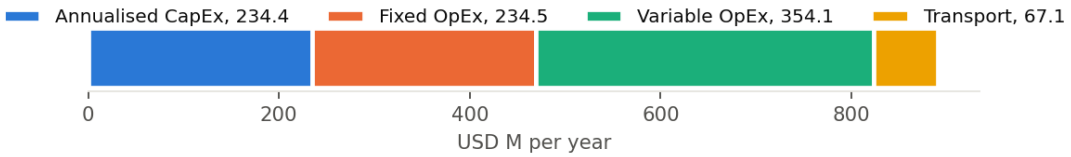


Figure 2. Annual cost by component, official base case.

Component	USD M/yr	Share
Annualised CapEx	234.4	26.3%
Fixed OpEx	234.5	26.3%
Variable OpEx, refineries	330.0	37.1%
Variable OpEx, depots	24.1	2.7%
Transport, site to depot	33.5	3.8%
Transport, refinery to airport	33.5	3.8%
Total	890.1	100%

Total CapEx is USD 4,689M, of which refineries are 97.9%. The objective excludes biomass purchase, as the update defines it, so USD 742/t is a delivered cost before feedstock. Readers should not compare it directly with a market SAF price.

11. Baseline comparison

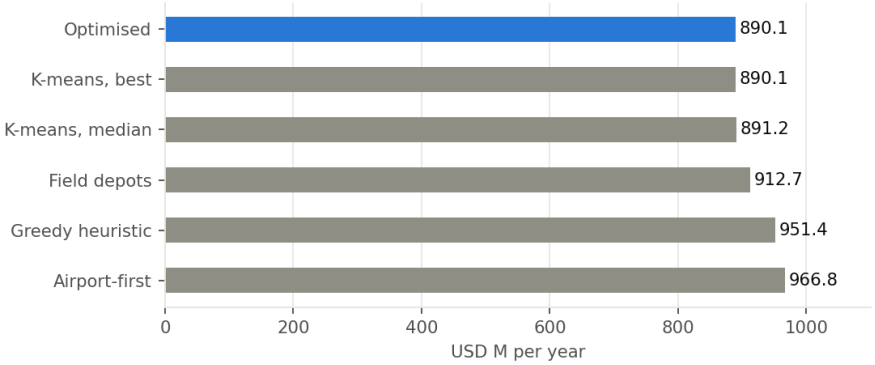


Figure 3. Annual cost on identical demand, distances and cost basis.

Design	USD M/yr	Extra cost vs optimised	Refineries	Depots	Truck-km, M/yr
Optimised	890.1	n/a	6	23	100.7
Field depots, 60 km radius	912.7	22.6 (2.5%)	6	25	75.3
Greedy: richest sites, farthest-first plants	951.4	61.3 (6.4%)	6	20	116.7
Airport-first: plants at PNK, PKN, BPN	966.8	76.7 (7.9%)	6	20	153.2
K-means, same fleet and refinement, best of 50	890.1	0.0 (0.0%)	6	23	n/a
K-means, same fleet and refinement, median	891.2	1.1 (0.1%)	6	23	n/a

All baselines passed the same audit. The greedy and airport rules are simple heuristics. K-means is deliberately strong, because it is handed the cheapest fleet, exact flows and our refinement. Its best run reaches our network.

That result cuts both ways. A separate starting method reaching the same cost is independent support for it. It also shows the site search adds little once the fleet is known, because transport is a small share of cost and the objective is flat near the optimum. The search still contributes the proven bound, the fleet and size choice when a scenario makes that unobvious, and the limits in section 13. These are model results, with no measured savings against real planners.

12. Depot trade-off

Measure	A: at refinery, optimised	B: in field, heuristic	Change
Annual total, USD M	890.09	912.66	+22.57
CapEx, USD M	4,689.0	4,695.8	+6.8
Annualised CapEx plus fixed OpEx, USD M/yr	468.9	469.6	+0.7
Variable OpEx, USD M/yr	354.1	354.3	+0.1
Logistics, USD M/yr	67.1	88.8	+21.7
Depots	23	25	+2
Refineries	6	6	+0
Biomass, kt/yr	6,000	6,000	-0
SAF, kt/yr	1,200	1,200	-0
Demand met, %	100	100	-0
Freight moved, billion tonne-km	0.754	0.897	+0.142
Estimated truck-km, millions	100.7	75.3	-25.4

Design B gathers biomass within 60 km and hauls the long leg in larger trucks, so it moves 18.8% more tonne-km but needs an estimated 25.2% fewer truck-km. That estimate assumes large trucks with empty returns and includes the barge portion to Tarakan, so it is not a count of vehicle trips. A tonne-km tariff cannot

see this benefit. B is a heuristic with no optimality proof, and it sits 3.18% above the proven bound.

13. Scenario and sensitivity analysis

Scenario	Status	USD M/yr	USD/t	Ref	Dep	Gap
Official base	feasible_gap	890.1	742	6	23	0.75%
High demand, +10%	feasible_gap	986.0	747	7	25	0.85%
High transport, +50%	feasible_gap	923.6	770	6	23	0.84%
All 6,800 kt, 340 kt per airport	feasible_gap	1015.1	746	7	25	2.66%
Field depots	heuristic	912.7	761	6	25	No proof
OSRM road distances	feasible_gap	917.4	765	6	23	0.60%
Real uplift at 5% blend	feasible_gap	50.0	10,031	1	1	0.43%

High demand forces a small plant into the fleet because six large plants take only 6,522 kt. Raising tariffs by 50% adds USD 33.5M and leaves the fleet at six plants. Figure 4 shows why: refinery terms dominate, and yield matters most, since it sets how much biomass and capacity are needed at all.

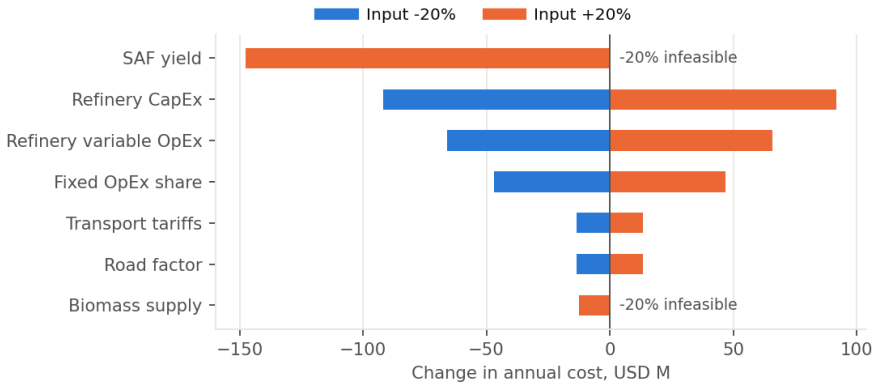


Figure 4. One-at-a-time changes of 20% around the base case. Yield and biomass at -20% are infeasible.

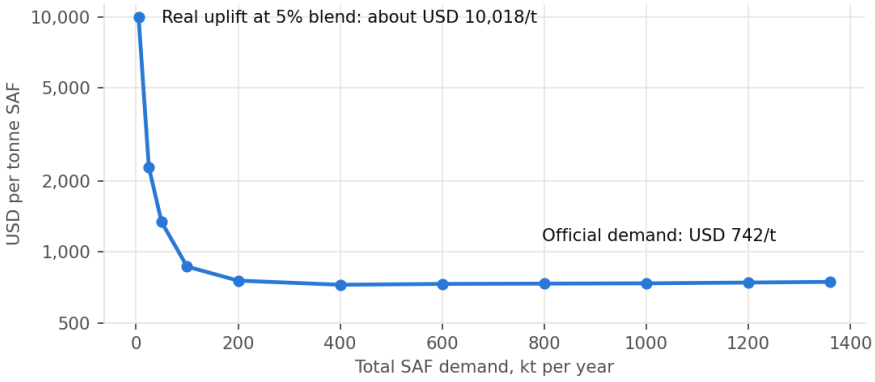


Figure 5. Cost per tonne by demand level. Each point is a full solve.

Marginal economics

Each row below is a full re-solve on candidate sites, against a base of USD 890.32M. These are finite-difference planning estimates and not shadow prices, because plant decisions are whole units.

Change	USD M/yr	USD per extra t	Fleet	Gap
Demand +10 kt	+4.8	481	6 large	0.69%
Demand +25 kt	+12.3	492	6 large	0.79%
Demand +50 kt	+25.0	499	6 large	1.55%
Demand +100 kt	+55.3	553	6 large	1.88%
Demand +104 kt	+57.8	556	6 large	1.90%
Demand +105 kt	+87.5	833	6 large, 1 small	0.71%
Demand +120 kt	+95.7	797	6 large, 1 small	0.85%
Demand +150 kt	+116.1	774	6 large, 1 small	2.22%
Large refinery capacity +100 kt	-29.2	n/a	5 large, 1 small	0.73%
Biomass supply -5%	+5.1	n/a	6 large	0.83%
Biomass supply -10%	+11.8	n/a	6 large	1.05%
Biomass supply -12%	Infeasible	n/a	n/a	n/a
Transport tariffs +10%	+6.7	n/a	6 large	0.67%
Transport tariffs +20%	+13.5	n/a	6 large	0.71%

Six large plants reach their limit at 1,304.4 kt of SAF. Below that, extra demand costs about USD 480 to 560 per tonne. The next kilotonne needs a seventh plant, so cost steps up sharply. Several increments are smaller than the solver gap in absolute terms, so small differences should not be over-read.

Reality check on demand

The official demand of 1,200 kt is far above what these airports use today. We estimate real jet uplift at about 100 kt per year, from Pertamina volumes at BPN and PNK [13, 14] and DGCA and BPS passenger counts [15], so official demand is about 12 times total current jet use. Indonesia's 1% SAF mandate from 2027 covers international departures from Jakarta and Bali only [16], so we applied Singapore's 3 to 5% target for 2030 [17] as a stand-in. A 5% blend gives about 5 kt SAF, which the model serves with one small refinery at 5% utilisation at USD 10,031/t. The official figure should therefore be read as a future or regional hub volume. Cost per tonne only settles once plants run near capacity, as Figure 5 shows.

Reality check on distance

With OSRM road distances in place of great-circle x 1.3, annual cost rises by USD 27.1M to USD 917.4M, a 3.0% change, and the fleet size holds. Transport's low share limits the effect.

Where the model fails

Case	Result	What the planner sees
Demand +20%	Infeasible	Demand of 1440 kt SAF needs 7200 kt biomass at yield 0.20, but only 6800 kt exists. This feedstock can make at most 1360 kt SAF. Cut demand by 5.6%, or raise yield to at least 0.212.
Yield 0.15	Infeasible	Demand of 1200 kt SAF needs 8000 kt biomass at yield 0.15, but only 6800 kt exists. This feedstock can make at most 1020 kt SAF. Cut demand by 15.0%, or raise yield to at least 0.176.
At most 5 refineries	Infeasible	5 large refineries take 5435 kt biomass, but this scenario needs 6000 kt. Allow at least 6 refineries.

Demand can rise by at most 13.3% before feedstock runs out. Yield below 0.176 makes 1,200 kt unreachable. The update allows yields of 15 to 25%, so the bottom of that range cannot meet official demand.

14. AI decision support

KASCO has two question layers. The Decision support tab is explainable and rule-based, with mathematical optimisation underneath. It uses no language model and no trained machine-learning model, which keeps every number reproducible. A question routes by rule to an intent, and each intent reads named fields from a

solver result and lists them under its answer. A refinery explanation includes a time-limited counterfactual solve with nearby sites closed. Questions about emissions, price or law are declined with a reason, because the model holds cost, mass and capacity only.

A what-if request is read clause by clause, so it can change several inputs at once, each with its own sign. It runs only when every part maps to a validated input, and otherwise it refuses and says which part it could not read. An audit found that the first version silently dropped a change and read the hyphen in 'what-if' as a minus sign. Both faults are fixed and covered by tests. The trade-off is that this layer understands a narrower range of phrasing than a language model would.

The second layer is an optional analyst panel that calls Claude Opus 5.5 directly from the browser. GitHub Pages cannot hold a private key, so the viewer pastes their own Anthropic key, which stays in the tab's memory and goes only to api.anthropic.com. The model receives a snapshot of solver results, can call the rule-based engine for what-ifs, and returns short bullets, practical actions, the visual to open and what to explore next. The app removes any line with a figure it cannot match to the snapshot, the question or the solver output. That check catches invented figures, but it cannot prove a matched figure supports the claim beside it, so a planner should confirm the answer on the visual it opens. A hosted version would need a proxy, an account and a spend cap, so that stays future work.

15. Validation and optimality evidence

A separate audit module recomputes every balance and the objective from flow records, using its own distance calculation. All 11 checks pass on the base case. Automated tests cover input data, capacities, mass balance, yield, demand, objective arithmetic, recalculation, infeasible cases, tampering and the what-if parser, and all 38 pass. Tampering tests confirm the audit fails when a flow or objective is altered.

As a second opinion, the Rust tool exports the co-located candidate-site model as a plain LP file and HiGHS 1.15.1 solves it separately. HiGHS reaches USD 890.3215M with a lower bound of USD 866.74M (2.65% gap at its 480 s limit). The Rust cost on the same candidate sites is USD 890.3215M with a bound of USD 883.60M (0.75% gap in 60 s). Two separate codes reached the same cost, which is strong evidence that the formulation and the search agree.

A tighter bound is only useful if it is valid, so we tested it on 5 smaller demand levels that HiGHS can close. In each case the Rust bound must not exceed the HiGHS optimum.

Total demand, kt	Rust cost	Rust bound	HiGHS cost	HiGHS bound	Bound valid
20	55.699	55.524	55.699	55.699	Yes
50	67.126	67.051	67.126	67.126	Yes
100	86.729	86.579	86.729	86.729	Yes
200	151.178	151.003	151.178	151.178	Yes
300	231.788	231.325	231.788	231.788	Yes

Status of each result: the base case is feasible within a proven gap on candidate sites, the five small cases match a proven optimum, refined coordinates are locally refined, and the field design is a heuristic. We make no claim of global optimality for the unrestricted location problem.

16. Limitations

The challenge's simplified inputs are necessary for a fair comparison, and they also shape the answer. Official demand is about 12 times our estimate of today's real jet use at these airports. The 80 biomass points are modelling inputs. A single tariff per leg hides road quality, river transport and seasonal access. Zero depot mass loss removes the main physical reason for field depots.

- The model is deterministic and annual, with no seasonality, storage or supply risk.
- Biomass price is outside the objective by instruction. Including it would favour the same sites only if prices are uniform.

- Co-located depots follow from zero mass loss and the published tariffs. Real wet feedstock could change this, as section 6 shows.
- The field design is a heuristic, so its USD 22.6M premium is an upper estimate.
- Site screening is preliminary. Land tenure, peat, forest status, water and hydrogen supply are unchecked, so no site is shown to be buildable.
- The land mask covers all of Borneo, so it does not separate Kalimantan from Sarawak and Sabah. All base-case plants lie in Kalimantan.
- The truck-km estimate assumes large trucks and empty returns, and it includes the barge portion to Tarakan.
- Real uplift for PKN and TRK is scaled from passengers, and the PNK volume is reported for West Kalimantan as a whole.

17. Future development

These are possible next steps, not commitments or achievements.

- **Secure language-model layer.** Natural-language requests would pass through a server-side proxy to structured parameters, the existing validator and the Rust solver. The model would never set costs, sites or results.
- **Real-world validation.** Replace the tariffs and distances with road, river and port data, then screen sites for land tenure, peat and forest status with local stakeholders [2].
- **Feedstock quality.** Replace the fixed yield with measured site data, including moisture and ash, as Li et al. recommend for logging residues [4]. Moisture alone decides whether depots belong in the field.
- **Uncertainty.** Once supply data can support an uncertainty set, test robust optimisation in the style of Keyvandarian et al. [27].

18. Conclusions

A six-plant network at USD 890M per year meets official demand, with the result checked by an independent audit and a second solver. The strongest levers are refinery cost and yield, and transport matters much less than a logistics framing suggests. KASCO's value lies in letting a planner test these questions in seconds, see where an idea fails and know how far an answer can be from the best possible. Given the right fleet, simple methods can get close, and we say so.

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